

M359 University conceptual data model

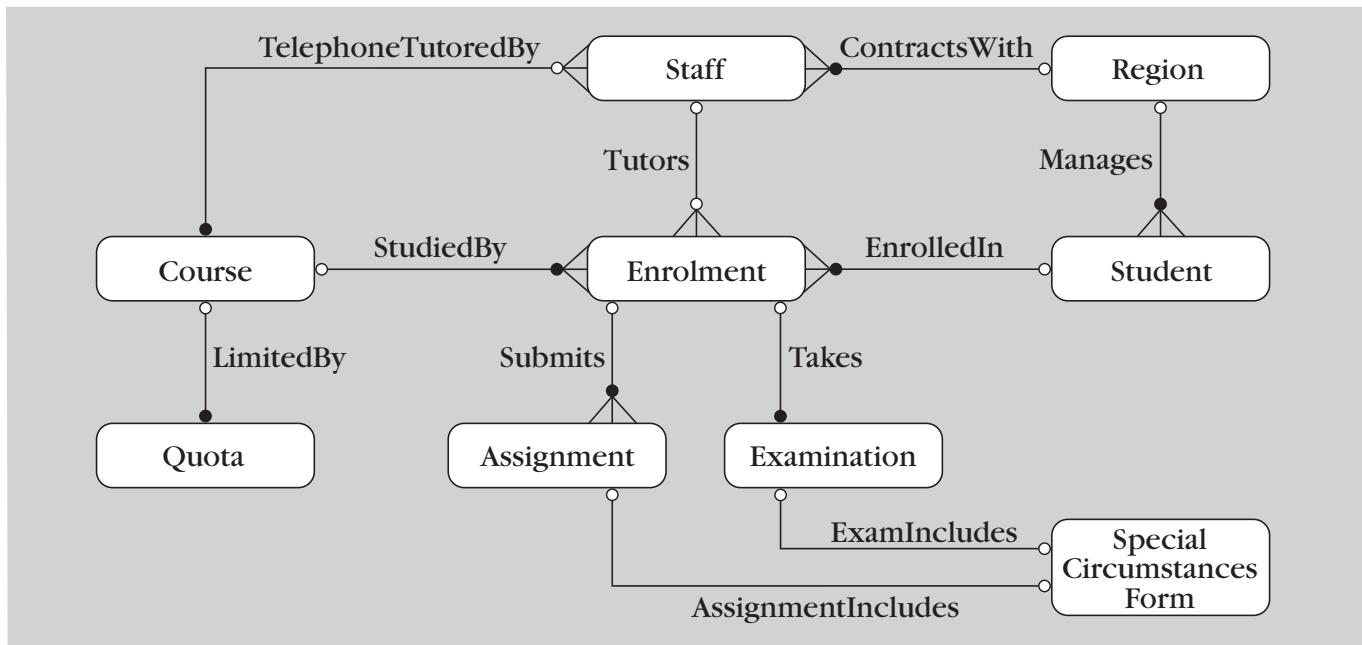


This conceptual data model has been developed from the University scenario example in *Block 1*, Section 5. The model is also used to support the course material in *Blocks 2* and *3*.

The details are supplied here so that you can see a complete conceptual data model for the University scenario.

This database card should be useful to you throughout the course.

Entity–relationship diagram



Entity types

Region (RegionNumber, Address, Telephone, EmailAddress)

Student (StudentId, Name, Address, EmailAddress, RegistrationDate)

Staff (StaffNumber, Name, Address, EmailAddress, Telephone)

Course (CourseCode, Title, Credit)

Quota (CourseCode, Limit, DateReviewed)

Enrolment (StudentId, CourseCode, EnrolmentDate)

Assignment (StudentId, CourseCode, AssignmentNumber, Mark)

Examination (StudentId, CourseCode, ExaminationLocation, Mark)

SpecialCircumstancesForm (FormNumber, DateReceived)

Additional constraints

- c.1 A staff member may only tutor students who are located in the region for which the member of staff has a contract. That is, an instance of the Staff entity type can only be related to an instance of the Enrolment entity type by the Tutors relationship if the instance of the Student entity type, to which that Enrolment instance is related by the EnrolledIn relationship, is related by the Manages relationship to the same instance of the Region entity type that the Staff instance is related to by the ContractsWith relationship.
- c.2 A staff member who acts as a telephone tutor must also tutor at least one enrolment on the same course. That is, an instance of the Staff entity type can only be related to an instance of the Course entity type by the TelephoneTutoredBy relationship if an instance of the Enrolment entity type, to which that Course instance is related by the StudiedBy relationship, is related by the Tutors relationship to that Staff instance.
- c.3 A student's registration date must be before the same student's enrolment date for any enrolment. That is, the value of the RegistrationDate attribute of the Student entity type instance must be less than the value of the EnrolmentDate attribute in the Enrolment entity type instance related to the Student entity type by the EnrolledIn relationship.
- c.4 Students may not enrol for more than 180 points' worth of courses at any one time. An instance of the Student entity type cannot be related to the Enrolment instances if the sum of the values of the Credit

attribute of the instances of the Course entity type to which the Enrolment instances are related by the StudiedBy relationship exceeds 180.

- c.5 The credit value of a course has a value of either 30 or 60 points. That is, the value of the Credit attribute of an instance of the Course entity type must have the value of either 30 or 60 points.
- c.6 A 60-point course may have up to five assignments and a 30-point course up to three assignments. That is, an instance of the Enrolment entity type cannot be involved in more than five instances of the Submits relationship if the value of the Credit attribute of the instance of the Course entity type to which the Enrolment is related by the StudiedBy relationship is 60. Furthermore, an instance of the Enrolment entity type cannot be involved in more than three instances of the Submits relationship if the value of the Credit attribute of the instance of the Course entity type to which the instance of the Enrolment entity type is related by the StudiedBy relationship is 30.
- c.7 Assessment marks have values that represent marks between 0 and 100. That is, the Mark attribute in instances of the Assignment and Examination entity types must take a value between 0 and 100.
- c.8 A special circumstances reference cannot be submitted for both an assignment and an examination, but must be submitted for one of these. That is, an instance of the SpecialCircumstancesForm entity type cannot be involved in both the ExamIncludes and AssignmentIncludes relationships, but must be involved in one of these relationships.
- c.9 The date a special circumstances form is received must be on or before today's date. That is, the value of the DateReceived attribute of an instance of the SpecialCircumstancesForm entity type must be on or before today's date.
- c.10 The date reviewed for a quota must be on or before today's date. That is, the value of the DateReviewed attribute for the Quota entity type must be on or before today's date.
- c.11 Quota is a weak entity type dependent on the entity type Course. So, the value of the CourseCode attribute in an instance of the entity type Quota must be the same value as the CourseCode attribute of the Course instance to which the entity type Quota is related by the relationship LimitedBy.
- c.12 Assignment is a weak entity type dependent on the entity type Enrolment. So, the value of the StudentId and CourseCode attribute pair in an instance of the entity type Assignment must be the same value as the StudentId and CourseCode attributes of the instance of the Enrolment entity type to which the entity type Assignment is related by the relationship Submits.
- c.13 Examination is a weak entity type dependent on the entity type Enrolment. So, the value of the StudentId and CourseCode attribute pair in an instance of the entity type Examination must be the same value as the StudentId and CourseCode attributes of the instance of the Enrolment entity type to which the entity type Examination is related by the relationship Takes.
- c.14 Enrolment is a weak entity type dependent on the entity type Student. So, the value of the StudentId attribute in an instance of the entity type Enrolment must be the same value as the StudentId attribute of the instance of the Student entity type to which the entity type Enrolment is related by the relationship EnrolledIn.
- c.15 Enrolment is a weak entity type dependent on the entity type Course. So, the value of the CourseCode attribute in an instance of the entity type Enrolment must be the same value as the CourseCode attribute of the instance of the Course entity type to which the entity type Enrolment is related by the relationship StudiedBy.
- c.16 The values for the region number of a region are limited to the range 1 through 12. That is, the value of the attribute RegionNumber in the Region entity type must come from the range 1 through 12 inclusive.

Assumptions (to be clarified with 'client' before completion)

- a.1 Students are not required to submit assignments in sequence and without omissions.
In this model, I have assumed that a student can submit Assignment 1, skip Assignment 2, then submit Assignment 3, and so on. Or they can submit Assignment 2 and then submit Assignment 1, followed by Assignment 3, and so on. Note that this is not a processing requirement; this determines if there is a constraint to enforce a limit such as 'Assignment number 2 cannot be recorded before assignment number 1', and so on.

Limitations

- l.1 Only current enrolments are of interest.
 - l.2 A student can have at most one enrolment on a specific course.
 - l.3 An assignment, for a specific enrolment, may be submitted only once.
 - l.4 An examination, for a specific enrolment, may be taken only once.
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